

ME2 - VCM on Raspberry Pi 5

Dataset-trained candidate: results, gates and Pi trial

Decision: The new intent model is installed as a separate Raspberry Pi trial, but the current model remains the default. On the single frozen test pass, the candidate scored slightly lower overall. It did better on smaller real/varied-source slices and had fewer out-of-scope actions at its own validation-selected gate. The Pi app is running, but live speech cannot be tested until a microphone is attached.

Why this dataset was not used earlier

The first pass audited the repulled files but did not fit a model because several source licenses and participant/voice permissions were not evidenced. The user then directed a private coursework training and comparison. Training/scoring ran locally; only model/runtime files were sent to the user's Pi, not raw dataset audio. Rights and consent still need confirmation before broader redistribution.

Training and evaluation

Source: ME2 Spoken Command Dataset, pinned revision a90b8d106349b02c5570a1a258503386043f63b2. Intent model: TinyDSCNN-48, 14,527 parameters, random initialization, no pretrained weights, selected epoch 82 of 100 on RTX 3050. Fit used 8,201 supported training clips and 2,122 source/speaker-group validation clips. The 66,390 numeral clips and 196-row holdout remained untouched.

The model, gates and scorer were frozen before the single paired evaluation of 4,418 published test clips: 4,371 supported commands and 47 true out-of-scope clips. The 158 unsupported-slot train rows were kept separate from true OOS. No wake-positive examples are present in this dataset, so the candidate retains the existing binary wake model byte-identically.

INT8 intent size: 39,315 bytes; paired wake+intent size: 76,740 bytes (74.9 KiB). Validation FP32/INT8 prediction agreement: 95.63%.

Paired test results

Model	Accuracy	Macro F1	Top-level accuracy	Top-level macro F1
Current intent model	79.64%	79.47%	81.38%	76.74%
New dataset candidate	78.95%	79.12%	82.25%	76.56%

Current leads aggregate accuracy and macro F1 by 0.69 and 0.35 percentage points. Candidate-minus-current top-level accuracy is +0.87 points; top-level macro F1 is -0.18 points. The source/speaker bootstrap intervals overlap, so this test does not establish an overall winner.

Accuracy by test source

Source slice	n	Current	Candidate
FluentSpeechCommands	147	22.45%	62.59%
SLURP	553	16.82%	26.04%
SNIPS	101	16.83%	32.67%
TimersAndSuch	2	100.00%	100.00%
group synthetic	3368	98.22%	92.84%
real voice	179	12.85%	24.02%
xela Multi-Sensor	11	18.18%	36.36%
xela SET TEMPERATURE REAL	10	30.00%	60.00%

Candidate performance is higher in several small real/varied-source slices and lower on group-synthetic speech (the largest slice). Source mix is imbalanced; use these values to plan live evaluation, not to claim a general real-world win.

Action gates and per-class results

Confidence and margin are accept/reject operating gates. They are not accuracy thresholds.

Model / gate	Confidence	Margin	Correct-action coverage	OOS false actions
Current deployed	0.68	0.15	77.30%	22 / 47
Current, validation-calibrated	0.98	0.00	65.41%	2 / 47
Candidate, validation-calibrated	0.76	0.00	61.36%	0 / 47

Candidate accepted-command accuracy: 98.82% at 61.36% correct-action coverage. It rejects more supported speech than the deployed current gate. The validation policy allowed at most 1% false actions on reject examples; only 27 such examples were available for gate selection.

Wake remains at threshold 0.95. On 4,418 wake-negative test clips, there was 1 false accept (0.02%). There are no wake positives, so wake recall cannot be measured here.

Per-class F1 (141 examples per label)

Label	Current F1	Candidate F1	Change (pp)
ALARM_6_00AM	86.02%	83.15%	-2.87
ALARM_8_00AM	72.58%	87.23%	+14.65
ALARM_9_00PM	76.38%	92.14%	+15.76
BRIGHTNESS_100	88.59%	89.89%	+1.30
BRIGHTNESS_20	92.42%	90.51%	-1.91
BRIGHTNESS_60	96.35%	96.35%	+0.00
CALL	67.55%	72.06%	+4.51
COLOR_BLUE	79.66%	66.41%	-13.25
COLOR_GREEN	80.15%	78.95%	-1.20
COLOR_RED	84.50%	71.48%	-13.02
CREATE_REMINDER_DRINK_WATER	82.57%	93.91%	+11.34
CREATE_REMINDER_EXERCISE	84.54%	92.47%	+7.93
CREATE_REMINDER_STUDY	96.06%	95.17%	-0.89
LIGHT_OFF	62.45%	49.80%	-12.65
LIGHT_ON	67.13%	63.87%	-3.26
LIST_REMINDERS	79.85%	80.60%	+0.74
MESSAGE	87.77%	83.69%	-4.08
NEXT	80.47%	72.35%	-8.11
PAUSE	75.94%	80.67%	+4.73
PLAY_MUSIC	55.41%	48.43%	-6.98
STOP	62.35%	67.31%	+4.96
TEMPERATURE_18	80.00%	91.03%	+11.03
TEMPERATURE_22	93.43%	87.20%	-6.23
TEMPERATURE_26	94.89%	92.96%	-1.93
TIME	70.97%	69.15%	-1.82
TIMER_10s	88.44%	94.29%	+5.85
TIMER_1m	89.49%	89.20%	-0.29
TIMER_30s	87.92%	93.23%	+5.31
VOLUME_DOWN	71.43%	59.52%	-11.90
VOLUME_UP	61.11%	54.62%	-6.49
WEATHER	67.26%	65.15%	-2.11

Only 2 of 31 candidate labels reach 95% F1. The per-class 95% target is unmet; the weakest candidate classes are PLAY_MUSIC (48.43%), LIGHT_OFF (49.80%), VOLUME_UP (54.62%), and COLOR_BLUE (66.41%).

Raspberry Pi trial

Installed and manually running at /home/dalmacio/Desktop/me2-dataset-candidate on cdfnjrpi5, bound to loopback port 7865. Assistant: <http://127.0.0.1:7865/assistant>. Autostart is off and GPIO is disabled, so the trial does not change the existing LED or default VCM state.

Check	Result
Bundle integrity	24 payload hashes passed
ARM64 ONNX CPU inference	passed
Wake frontend + model p95	5.510 ms
Intent frontend + model p95	5.225 ms
Loaded model hashes	match frozen candidate and retained wake
API and Assistant page	HTTP 200
Microphone	No microphone connected

The Pi app and CPU models are running, but the Pi reports no microphone and zero active capture. Do not count this as a live speech test. The Windows SSH tunnel currently makes the candidate UI available at 127.0.0.1:7865.

Next step

Connect a microphone, then compare both model versions with the same speaker, microphone placement, exact prompts, room noise, and randomized order. Record wake hits/false wakes per hour, intent actions/rejects, and lights-off/color mistakes. Keep the current release as default unless the personal trial shows a useful gain without worse OOS rejection. Do not tune against the already-scored frozen test.

Evidence: frozen_test_comparison.json, candidate training/quantization outputs in runs/classagreementvcm-20261002-a90b8d10-r3, and deployment/classagreementvcm_candidate_deployment.json. The full Markdown report includes interpretation and the complete source-slice table.